

# Wang Yao (王曜)

## Postdoctoral Research Fellow

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ORCID [[link](#)] | Google Scholar [[link](#)]



## Professional experience

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**Nanyang Technological University**, Postdoctoral Research Fellow

*Feb 2026-Now*

- **Supervisor:** Associate Prof. Xunchang Fei

## Education

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**Nanyang Technological University**, Ph.D. in Environmental Engineering

*Aug 2022–Jan 2026*

- **Supervisor:** Associate Prof. Xunchang Fei
- **Advisory Committee:** Prof. Lim Teik Thye, Associate Prof. Grzegorz Lisak
- **Thesis:** The biogeochemical dynamics of major elements in municipal solid waste landfills

**Nanyang Technological University**, M.Eng. in Environmental engineering

*Aug 2020–Jul 2022*

- **Supervisor:** Associate Prof. Xunchang Fei
- **Thesis:** The degradation processes of municipal solid waste in landfills

**Tongji University**, B.S. in Environmental science

*Sep 2016–Jun 2020*

- **Supervisor:** Prof. Daqiang Yin, Prof. Hongtao Wang
- **Thesis:** QSAR modeling of chlorinated paraffin in aquatic ecosystems

## Research interests

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My research focuses on advancing sustainable waste management by understanding the cradle-to-grave fate of waste, with the goal of reducing carbon emissions and pollution from waste management system. My current main research topics include:

### AI-driven decision-making for sustainable waste management.

I develop a local-to-global waste inventory using interpretable machine learning, enabling more detailed scenario planning and policy evaluation from national down to city scales. Future research will extend to satellite-based detection of global dumpsites and waste management facilities, aiming to deliver high-resolution assessments of human health risks.

*Related publication(s):* [FA-2](#), [FA-5](#), [CA-6](#). Additional manuscripts in preparation.

### Greenhouse-gas accounting and MRV for waste systems.

I develop bottom-up models tailored to local conditions and reconcile them with observations to correct biases in default inventories. The results reveal underestimation hotspots and prioritize cost-effective mitigation portfolios. Future research will focus on elucidating the fundamental mechanisms of GHG emissions, with the goal of enabling more targeted control strategies.

*Related publication(s):* [FA-3](#), [FA-4](#), [CA-2](#), [CA-3](#), [CA-1](#).

## Climate–waste interactions and resilience of land-disposed waste.

I quantify element biogeochemical dynamics in landfills and diagnose how climate variability and extremes alter the emission / leakage pathways. The findings translate into design principles for climate-resilient waste infrastructure. Future work will specify site-specific climate change impact on waste management facilities.

Related publication(s): [FA-1](#), [CA-5](#), [CA-11](#).

## Fates of plastics and microplastics in disposal sites.

I investigate the behavior, fragmentation and degradation mechanisms, transport, and environmental leakage of plastics in waste management systems. The results inform monitoring protocols, risk assessment, and end-of-life strategies. Future research will focus on the formation mechanisms, characteristics, and environmental impacts of plastiglomerates, a unique plastic–rock composite that can develop under landfill conditions.

Related publication(s): [FA-6](#), [CA-4](#), [CA-8](#), [CA-9](#), [CA-10](#).

## Publications

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Google Scholar Total citations = 390, H-index = 9, Average IF per paper = 21.8      Latest update: Mar/2026

### First author

- FA-1** Yao Wang, Zhou, C., Zhang, H., Ma, S., Lou, Z., Raskin, L., Skerlos, S., Coulon, F., Zekkos, D., Hussain, A., Hussain, A., Lisak, G., Fang, M., Yin, K., He, H., Yuan Wang, Fei, X.\* , 2025. Biogeochemical dynamics of major elements in municipal solid waste landfills can induce health risks for nearly 1 billion people. **One Earth** (IF=16.1), 8(9), 101418. <https://doi.org/10.1016/j.oneear.2025.101418> (Citations: 4)
- FA-2** Wang, Y., Yin, K., Fang, M., Fei, X.\* , 2025. Underappreciated majority of solid waste: insufficient data and unclear impacts. **The Innovation** (IF=25.7), 6(3), 100776. <https://doi.org/10.1016/j.xinn.2024.100776> (Citations: 4)
- FA-3** Wang, Y., Fang, M., Guo, Y., Pi, X., Wang, Y.J., He, H., Lou, Z., Yin, K.\* , Fei, X.\* , 2024. Methane emissions differentially underestimated from landfills worldwide. **Nature Sustainability** (IF=29.2), 7, pp.496–507. <https://doi.org/10.1038/s41893-024-01307-9> (Citations: 103, ESI Highly Cited Paper).
- FA-4** Wang, Y., Zhou, C., Lou, Z., Zhang, H., Hussain, A., Zhan, L., Yin, K., Fang, M., Fei, X.\* , 2024. Underestimated methane emission from solid waste disposal sites reveals missed greenhouse gas mitigation opportunities. **Engineering** (IF=10.1), 36, pp.12-15. <https://doi.org/10.1016/j.eng.2023.12.011> (Citations: 17)
- FA-5** Wang, Y., Ruiz-Acevedo, A., Rameez, E., Raghavan, V., Hussain, A.\* and Fei, X.\* , 2024. Toward sustainable waste management in small islands developing states: integrated waste-to-energy solutions in Maldives context. **Frontiers of Environmental Science & Engineering** (IF=6.7), 18(2), p.24-37. <https://doi.org/10.1007/s11783-024-1784-7> (Citations: 11)
- FA-6** Wang, Y., Lu, X. and Fei, X.\* , 2021. Property changes of conventional plastic waste mixed with municipal solid waste after 10-year degradation experiments simulating landfill conditions. **Journal of Hazardous Materials Letters** (IF=6.6), 2, p.100047. <https://doi.org/10.1016/j.hazl.2021.100047> (Citations: 22)

### Contributing author

- CA-1** Yin, K.\* , Zhu, J., Wu, M., Zhang, H., Ling, X., Cai, M., Ren, S., Zhao, X., Ling, C., Yu, L., Tong, H., He, C., Wang, Y., and Fei, X.\* (2026). Misbehaviour dominates GHG emissions from food loss and waste. **Nature Climate Change** (IF=30.3). <https://doi.org/10.1038/s41558-026-02596-y> (Citations: NA)
- CA-2** Ye, R., Yin, K., Yang, Z., Zhang, C., Huo, W., Wang, Y., Zhang, H.\* , Fei, X.\* , 2026. A Novel Semi-Empirical Model for Multi-Scenario Estimations of Nitrous Oxide Emissions from Municipal Solid Waste Landfills. **Environmental Science & Technology** (IF=11.3), 60, 1, 595–608. <https://doi.org/10.1021/acs.est.5c08871> (Citations: NA)
- CA-3** Yang, Z., Yin, K., Ye, R., Zhang, C., Wang, Y., Zhang, H.\* , Fei, X.\* , 2025. Revealing hidden methane mitigation opportunities from legacy dumping and landfilling of municipal solid waste in rural China. **Journal of Environmental Management** (IF=8.4), 394, 127244. <https://doi.org/10.1016/j.jenvman.2025.127244> (Citations: 1)

- CA-4 Yadav, V.\*, Fei, X., Arora, M., van Emmerik, T.H., **Wang, Y.** and Laurent, A., 2025. Gaps in quantifying environmental losses of plastics impede effective solutions. **Nature Reviews Materials** (IF=79.8), pp.1-3. <https://doi.org/10.1038/s41578-025-00802-5> (Citations: 2)
- CA-5 Wang, Y., Pi, X., Yadav, V., Hussain, A., Liu, Q., **Wang, Y.**, Guo, Y., Zhang, Y. and Fei, X.\*, 2025. Leaching model of an expanding coastal dumpsite considering climate change. **Waste Management** (IF=8.1), 200, 114772. <https://doi.org/10.1016/j.wasman.2025.114772> (Citations: NA)
- CA-6 Zhang, Z., Wang, J., Li, J., **Wang, Y.**, Yin, K. and Fei, X.\*, 2024. Impacts of regional socioeconomic statuses and global events on solid waste research reflected in six waste-focused journals. **Waste Management** (IF=8.1), 182, pp.113-123. <https://doi.org/10.1016/j.wasman.2024.04.028> (Citations: 5)
- CA-7 Pi, X., Fei, X.\*, **Wang, Y.**, Sun, X. and Guo, Y., 2023. Global void ratio of municipal solid waste for compression indices estimation. **Waste Management** (IF=8.1), 160, pp.69-79. <https://doi.org/10.1016/j.wasman.2023.02.003> (Citations: 12)
- CA-8 Lu, X., He, H., **Wang, Y.**, Guo, Y. and Fei, X.\*, 2023. Masses and size distributions of mechanically fragmented microplastics from LDPE and EPS under simulated landfill conditions. **Journal of Hazardous Materials** (IF=13.6), 445, 130542. <https://doi.org/10.1016/j.jhazmat.2022.130542> (Citations: 26)
- CA-9 Fei, X.\*, He, H., Pi, X., Lu, X., Chen, Q., Ma, J., **Wang, Y.**, Fang, M., Wu, C. and Feng, S., 2023. The distribution, behavior, and release of macro-and micro-size plastic wastes in solid waste disposal sites. **Critical Reviews in Environmental Science and Technology** (IF=12.6), 53(3), pp.366-389. <https://doi.org/10.1080/10643389.2022.2054649> (Citations: 46)
- CA-10 Fei, X.\*, Guo, Y., **Wang, Y.**, Fang, M., Yin, K. and He, H., 2022. The long-term fates of land-disposed plastic waste. **Nature Reviews Earth & Environment** (IF=71.5), 3(11), pp.733-735. <https://doi.org/10.1038/s43017-022-00354-0> (Citations: 53)
- CA-11 Fei, X.\*, Fang, M. and **Wang, Y.**, 2021. Climate change affects land-disposed waste. **Nature Climate Change** (IF=30.3), 11(12), pp.1004-1005. <https://doi.org/10.1038/s41558-021-01220-5> (Citations: 81)

Note: \* indicates the corresponding author. Full list of publications on:

ORCID: <https://orcid.org/0000-0003-2373-0201>

Google Scholar: <https://scholar.google.com/citations?hl=en&user=aTPoRrMAAAAJ>.

## Grants

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- **Participant**, "Breaking the ground" in Semakau Landfill: technological solutions for site investigation and material reuse, Singapore National Environment Agency (NEA), 3,000,000 SGD (2,400,000 USD)
- **Participant**, Long-term environmental behaviour for treated incineration bottom ash, Singapore National Environment Agency (NEA), 6,000,000 SGD (4,800,000 USD)
- **Participant**, Tackling microfibres at source: investigating opportunities to reduce microfibre pollution from the fashion industry through textile design and manufacturing innovation, United Nations Development Program (UNDP), 56,250 USD

## Service

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### Editorial role

- Young Editorial Board Member, *The Innovation Geoscience* [\[link\]](#)
- Young Editorial Board Member, *DeCarbon* [\[link\]](#)

### Reviewer

- Nature Communications
- Journal of Environmental Management
- 9th International Congress on Environmental Geotechnics (ICEG)

### Talk

- 2024 Frontiers of Environmental Science & Engineering (FESE) Young Talent Forum, Tsinghua University

### Teaching

- CV0003 Introduction to Data Science and Artificial Intelligence, NTU, Y24/25, Semester 2  
Led tutorials on Python programming, data analysis, and machine learning fundamentals.
- CV1014 Introduction to Computational Thinking, NTU, Y24/25, Semester 1  
Led tutorials on algorithmic thinking and introductory programming.
- EG0001 Engineers & Society, NTU, Y23/24, Semester 1 & 2  
Facilitated discussions on engineering ethics, and supported assessment and feedback.

### Student supervision

- Koh Xin Li, NTU, Environmental Engineering, Year 4, Final Year Project: Climate change effects on landfills.
- Steffi Rei Wong Shi Qi, NTU, Environmental Engineering, Year 4, Final Year Project: Carbon emissions from solid waste in Southeast Asia.
- Sim Zu Da Joel, NTU, Civil Engineering, Year 4, Final Year Project: Estimating landfill carbon stocks and emissions.
- Hou Tianyuan, NTU, Environmental Engineering, Year 4, Final Year Project: GHG emissions from waste sector.
- Soo Wei Song, NTU, Environmental Engineering, Year 4, Final Year Project: Method refinement for estimating carbon stock in landfills

### Scholarships & Awards

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- **Frontiers Planet Prize, 2026**  
Selected as Singapore's sole National Champion and one of 25 National Champions worldwide for a sole-first-author *Nature Sustainability* paper on landfill methane emissions. The Frontiers Planet Prize is a major global scientific award described as “the world’s biggest prize in scientific solutions for planetary health”.
- **Chinese Government Award for Outstanding Self-Financed Students Abroad, 2025**  
A prestigious scholarship established by the Chinese government to recognize outstanding non-CSC funded Chinese students studying abroad. Only 650 recipients are selected worldwide each year for their exceptional academic and research achievements.
- **Nanyang Technological University Research Scholarship, 2022–2026**  
A full scholarship awarded to outstanding Ph.D. students, covering full tuition fees and providing a monthly living allowance.
- **Nanyang Technological University Research Scholarship, 2020–2022**  
A full scholarship awarded to outstanding Master's students, covering full tuition fees and providing a monthly living allowance.
- **NTU CEE Best Research Poster Award (Bronze), 2025**
- **Tongji Scholarship of Excellence**, Tongji University, Academic Years 2016–2017, 2017–2018, 2018–2019